Legible Text

3

An optometrist supposedly posted the following sign in her office, "If you don't see what you are looking for, you've come to the right place." But not being able to read text easily in a PowerPoint presentation is no laughing matter; in fact, it might give the audience the opposite sign, that they've come to the wrong place.

Virtually all presentations include text, and illegible text is a major cause of many confusing PowerPoint presentations. In this chapter you'll find recommendations for how to create clear and legible bulleted lists, labels, titles, tables, and keys.

Let's start with the most basic element of all: what font you use for your text.

Ten Tips on Fonts

To use the immortal words of President George W. Bush, "If we don't succeed, we run the risk of failure." There is no easier way to fail than to ensure that the audience cannot read the words on your slides. In this section, I present recommendations to how to avoid the risk of failure because the audience sees only smudges instead of words, or has to strain to decode your text. Here we will focus on fonts. The term "font" refers here to the general appearance of letters; fonts vary in terms of their typeface (such as Times or Ariel), style (such as *italics*, UPPERCASE, or **bold** vs. normal weight), size, and color. Many fonts are available for the PowerPoint program, and to some degree your choice will be dictated by your personal taste. However, some fonts are definitely better than others for presentations.

1. Avoid all uppercase, all italics, or all bold.

Fonts in which letters share many features are difficult to read because it is not easy to separate one letter from the next, so the viewer must look carefully at each letter.

> **Example**: compare abcdefghijklmnop versus ABCDEFGHIJKLMNOP. The forms of the letters in the second font are more similar than in the first.

- Use mixed uppercase and lowercase letters; don't use all uppercase letters, which are much more alike than a mixture of upper- and lowercase letters (and also, who wants to be shouted at?).
- Use standard font whenever possible; don't use italics for more than a word or two.
- Use normal weight as the default, and don't use bold throughout; letters in bold are less discriminable than normal weight letters, and also are too salient.

[Don't]

DON’T USE ALL UPPERCASE
and don’t use all italics

[Do]

Use a mixture of uppercase and lowercase instead

2. Don’t underline.

Underlining cuts off the bottoms of letters that have descending lines (such as p and g), which makes them harder to identify. If you want to emphasize a word, put it in italics (provided that the letters are still easy to distinguish in that font), make it bold, or use a more salient color.

[Don't]

Don’t underline

[Do]

Use **bold,** *italics,* or a change in color for emphasis

3. Use different colors only for emphasis or to specify different classes of information.

You’re likely to notice the nail that sticks up above the boards because attention is drawn to large differences. Like that nail, changes in color draw attention, and we expect changes in color to

signal changes in content. So don't change the color of the font unless you want to emphasize something or distinguish among different classes of information. (For color versions of the examples below, see insert, Figures 3.3a and 3.3b.)

- When using color to signal a change, use a color for only one term or phrase. If you make more than one word the same color, know

[Don't]

Don't vary color purely for decorative purposes

[Do]

Vary color only for emphasis or to group words together

that readers will tend to group together words in the same color—even if you are only using the color for emphasis and the words are not in fact closely related.

- Avoid using more than three different colors; too many colors can easily produce too many perceptual units and overwhelm the viewers' short-term memory capacity.

4. Use different fonts only for emphasis or to specify different classes of information.

Vary the font only to emphasize or group material into distinct classes.

5. Use fonts that are easy to read.

Editorial style manuals often urge us not to use a twenty-five-cent word when an easier-to-understand nickel one will do. The simpler-is-better rule also goes for fonts. At first blush, you might think that a fancy font will make you look sophisticated. Not so. You may see the font as fancy, but the audience may see it as an obstruction. Instead of making you look good, an ornate font will make the

[Don't]

Don't vary font purely for decorative *purposes*

[Do]

Vary font only for *emphasis* or to group words together

audience work harder. Letters in such fonts take more time and energy to read because the viewer must search for the relevant lines and try to shut out the flourishes. Even worse, such letters may also clump together visually, making the words difficult to distinguish.

[Don't]

Don't use visually complex fonts

[Do]

Use visually simple fonts

6. Ensure that words are large enough to read easily.

Text must be clearly discernible even to the viewers sitting at the very back of the room. To get a sense of their viewpoint, stand about three feet from your computer screen. If you can read the text on it without effort, it's probably large enough for your presentation. But the key is not the actual size of the letters on your computer screen—or even their size measured on the screen during a presentation. Instead, the key is the visual extent of the letters, as seen from a specific distance: As you move farther from the screen, the letters occupy an increasingly smaller visual extent (think of taking a photo of the screen from different distances—the size of the text in the photo will be increasingly smaller as you move increasingly far from the screen). More precisely, hold up your thumb at arm's length and look at it through one eye. The optimal size for letters is a bit less than one-quarter of the visual extent of your thumb as it appears at arm's length.

7. Ensure that viewers can easily discriminate text from the background.

The PowerPoint program lets you use many dramatic and interesting background patterns. But before becoming too enamored with those celestial scenes or dramatic splashes of color, check to make sure that the background isn't so salient, or (worse) similar to the text, that it's difficult to read the information-conveying words.

[Don't]

Ensure that the font is large enough to be seen from the back of the room

[Do]

Ensure that the font is large enough to be seen from the back of the room

- If you want to use such backgrounds, lower their contrast to make them, well, "fade into the background," letting your text assert itself in the foreground (see Figure 3.7b).
- To be on the safe side, use a background with low contrast and an even color, and a font color that is easily discriminated.

Example: An orange-yellow or gold font on a dark gray background is highly discriminable, as is a white font on any highly saturated "cool" color, or a black font on a very unsaturated, pale color. (See Chapter 4 for more detail.)

8. Use either a serif or sans serif font, but don't mix and mingle them arbitrarily.

Typefaces are divided into two general classes, *serif* and *sans serif*. Serif typefaces (such as Times and Palatino) have little feet, brackets and hooks at the ends of the lines used in the letters; sans serif typefaces (such as Ariel and Century Gothic) have only straightforward strokes. Use the same font consistently—otherwise changes will be taken to convey information, and ***the*** text may ***look*** `very strange` and is harder to read.

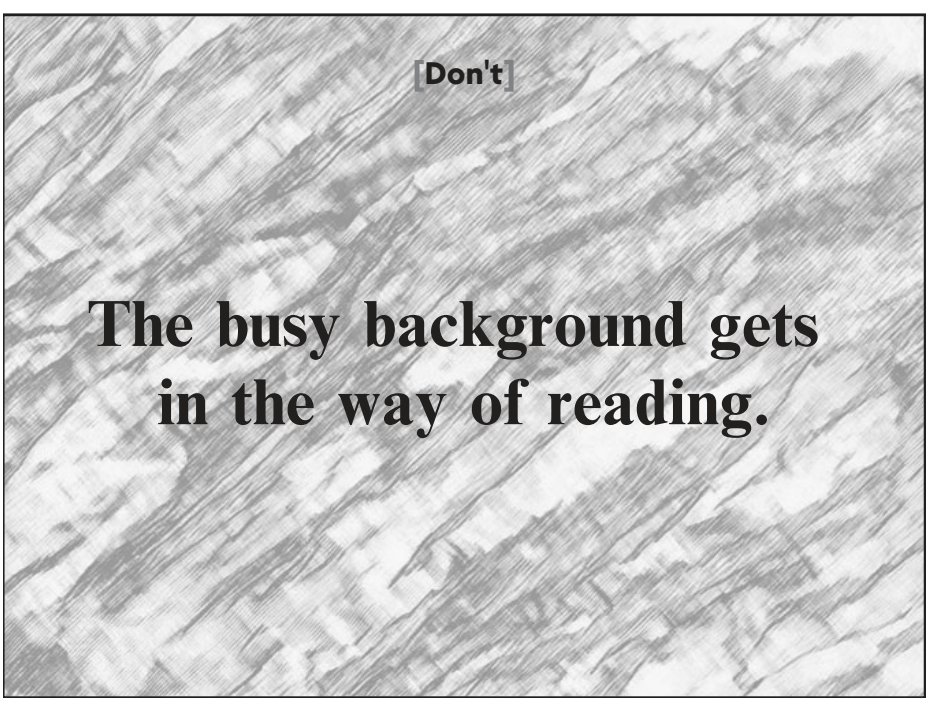

Figure 3.7a. If the background is too salient, viewers will have to work to discriminate text from the background. A display should not be a haystack, with your message the needle. If you use a patterned background, make sure that it does not bury the text and that it is not so salient as to distract from your message. (See color version in insert.)

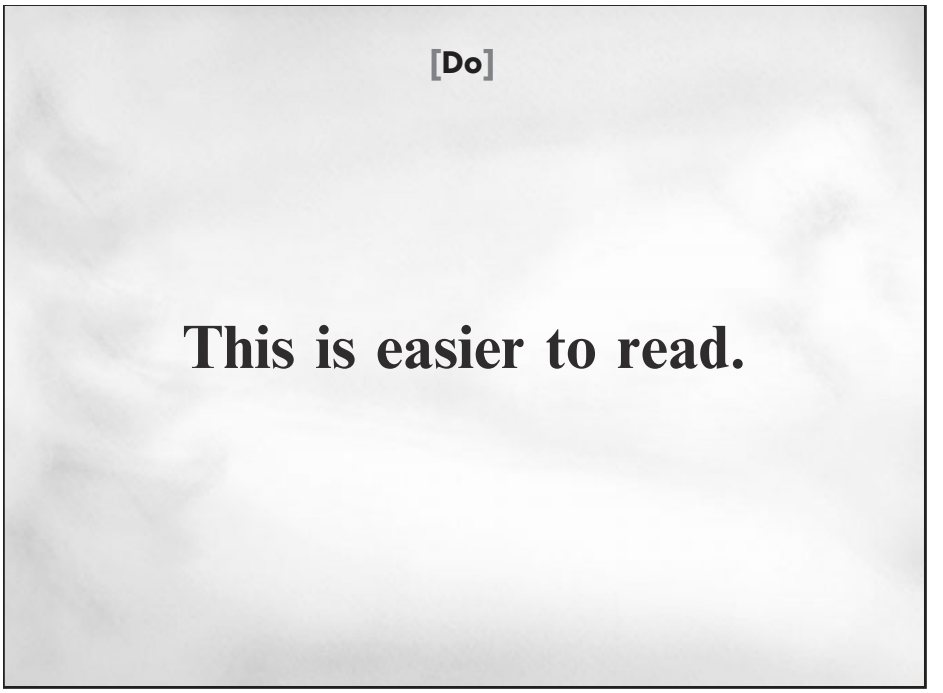

Figure 3.7b. (For color version, see insert.)

- If fonts are so small that they are barely legible, then sans serif is better because in a serif font, the little feet, brackets, and hooks in different letters become inappropriately grouped together, making the letters hard to distinguish.
- If the contrast or luminance (which our perceptual systems register as brightness) is so low that viewers can barely see the type, then serif fonts are better because the aforementioned little feet, brackets, and hooks provide additional cues regarding the identity of each letter.

That said, if these variables are important, reconsider your slides! You should never put the audience in the position of having to read words that are too small or hard to see! The bottom line: If you present text at a large enough size and in conditions of sufficient contrast and illumination, then any standard (such as Ariel or Times, not ornate or "fancy") typeface is acceptable.

9. Use standard fonts.

Unfortunately, fonts do not always appear the same when you make your actual presentation as they did on your computer screen. Not all

[Don't]

If fonts are very small, so that they are barely legible, use sans serif font; if the contrast or luminance is low, so that fonts that should be visible can barely be seen, use serif fonts.

[Do]

If the font is large enough and visible enough, either serif or sans serif font is acceptable

computers are loaded with the same font set, and some projectors have low resolution, which can also lead to illegible type. To be safe, use serif fonts such as Times New Roman (the most common), Garamond, or Century Schoolbook, or sans serif fonts such as Arial, Verdana, or Tahoma. I used to use only Palatino on the Mac, but I discovered that some computers don't have this font and will substi-

tute Times New Roman, which changes the presentation's spacing. Printers will also substitute for fonts they don't recognize, so don't count on your printouts looking the same if you use an uncommon font. To be on the safe side, use TrueType™ fonts, and an XGA (1024 × 768 resolution) projector. To quote President Bush again, "One word sums up properly the responsibility of any governor, and that one word is `to be prepared.' "

[Don't]

Don't use non-standard fonts

[Do]

Use standard fonts

10. Ensure that only the relevant words are distinguishable from the background.

I recently watched a DVD of a concert by the rock band the Who. Fourteen musicians were on stage, but the spotlights picked out only the three (then) surviving members of the original band. Without the spotlights, it would have been easy to be distracted by the new lead guitar player, or that attractive woman singing backups. You need to supply the spotlight for your audience, leading them to pay attention to just the part of the slide that's immediately relevant. To ensure that the audience members' attention will not wander, gray out all parts of the slide but the section you are discussing (see Figure 3.10b). As noted earlier, by "gray out," I mean switch the color or density so that the font is just barely distinguishable from the background. When you gray out an irrelevant part of your presentation, the viewer will be aware that something is there, but not be tempted to attend to it. Present each new section when it becomes relevant by changing its contrast to the background.

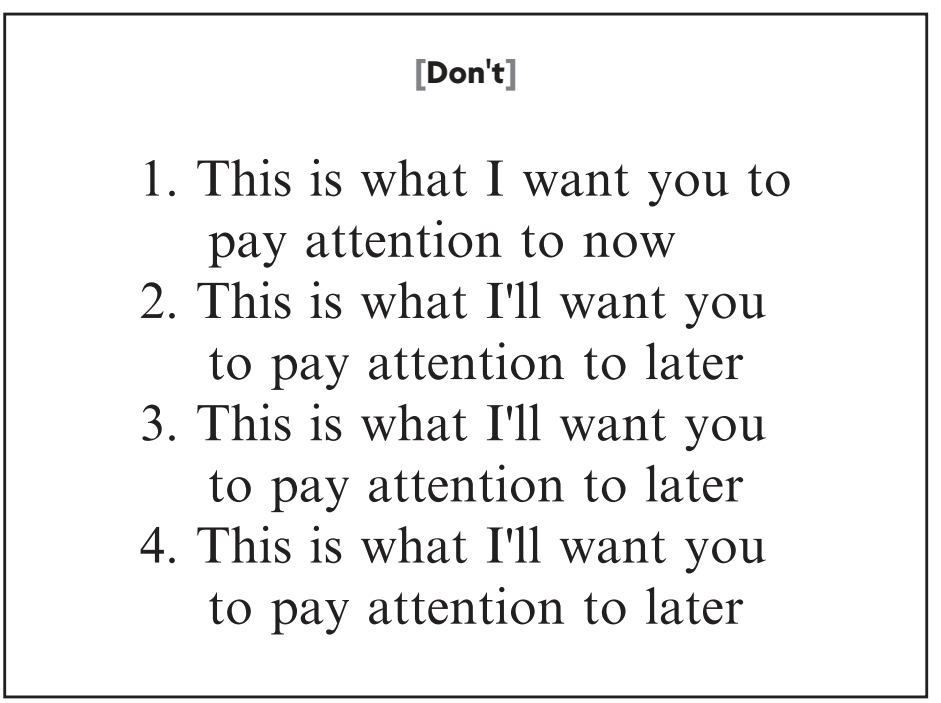

Figure 3.10a. Use graying out to direct the audience's attention to what is relevant at a particular point in the presentation.

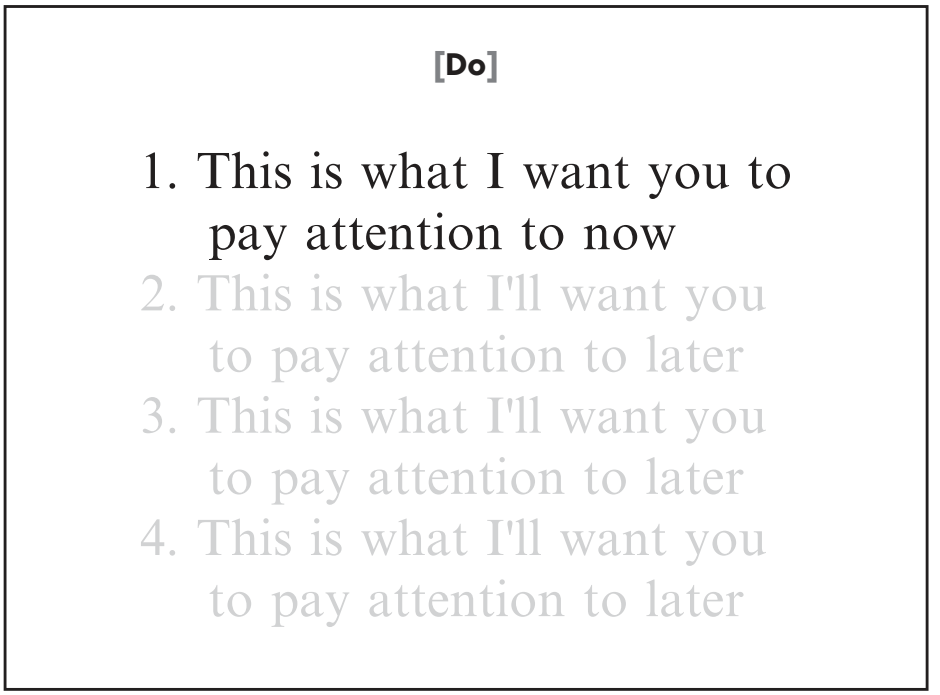

Figure 3.10b.

Bullets: Guidelines for Straight Shooting

Bullets are a convenient way to present the separate elements of a list. However, bullets are a bit like salt—often essential to bring out the best, but distasteful if overdone.

1. Present a quick overview of a list, then gray out all but the first entry.

Begin by presenting a complete list of bulleted items, but only long enough for the viewers to see the length of the list. Ensure that the audience pays attention only to the first item on your list (which will be salient) by graying out everything but this item, so that the remainder of the list can be seen, but not easily read.

2. Present items individually, highlighting only one at a time as you go from top to bottom.

Finish reading and discussing each item before you present the next. Gray out all entries other than the one you are presenting, including

the previous one, which will ensure that you direct the viewers' attention to just the material you want them to focus on.

3. Use bullets only for topic sentences or specific cases.

Bulleted entries on a list should provide only key concepts or examples. Don't present every word in your entire presentation in bulleted lists: Present no more and no less than you need to help the viewers organize and understand your message.

Example: Present only enough information in an opening outline to orient the viewers (the panel shown below, Figure 3.11a, would be overwhelming, even in the best of circumstances).

4. Give the audience time to read each bulleted item.

Don't be caught saying one thing while the audience is trying to read another.

[Don't]

I. The Three Goals
  A. Connect with your audience
  B. Direct and hold attention
  C. Promote understanding and memory
II. Specific Principles to Help Achieve Goals
  A. Connect with your audience
    1. Relevance
    2. Appropriate knowledge
  B. Direct and hold attention
    1. Salience
    2. Discriminability
    3. Perceptual organization
  C. Promote understanding and memory
    1. Compatibility
    2. Informative changes
    3. Capacity limitations
III. The Good, the Bad, and the Incomprehensible

Figure 3.11a. An opening outline should provide a general road map for the presentation. The material above is not only overwhelming, but much of it won't mean anything to the audience until they've heard later parts of the presentation.

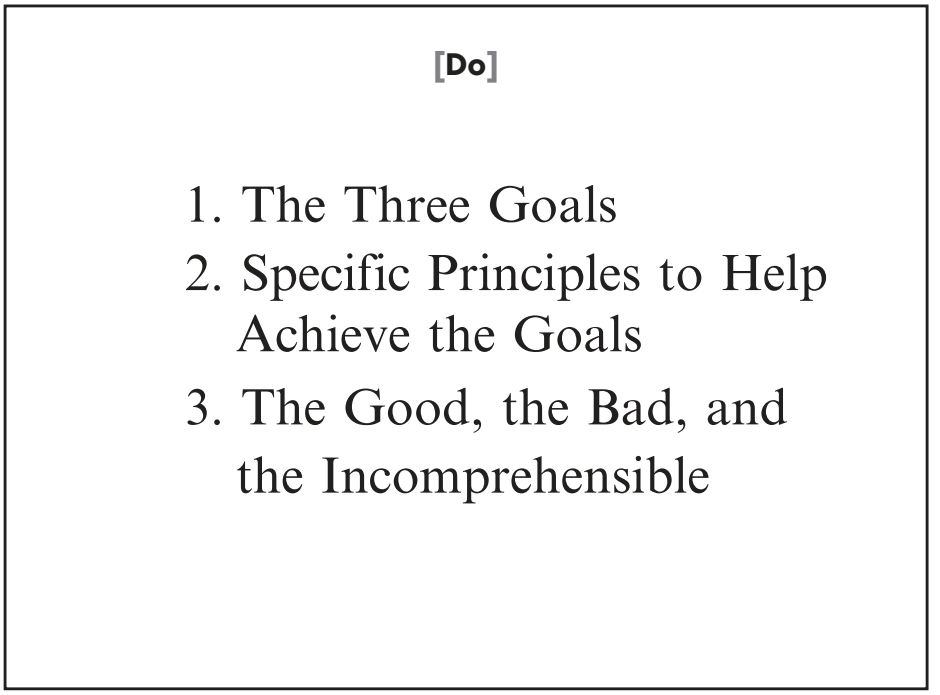

Figure 3.11b.

5. Discuss only material closely related to the current entry.

After you read a bulleted item, you can expand on the concept, such as by paraphrasing it or providing an anecdote that illustrates it. But stay close to that concept. If you present a new idea, use another bullet. In general, discuss only material that is closely related to what is currently being shown on a slide.

6. Use no more than two lines per bulleted entry.

As a general rule, include no more than two lines per entry in a bulleted list. Two lines are roughly enough space for four concepts, which—as codified in our *Rule of Four*—is all that viewers can reflect on easily at any one time. A concept corresponds to a word or phrase that expresses an idea. A simple rule of thumb for identifying the number of concepts is to count the nouns and verbs. For example, "Use two lines per bulleted entry" includes three concepts: "Use," "lines," and "entry." (In fact, concepts usually can be decomposed into groups of concepts, such as "two lines" and "bulleted entry"—but these combinations serve as single groups, which is all that is important for present purposes.)

[Don't]

This presentation will focus on the psychological principles that make a PowerPoint presentation clear and persuasive and will explain how to use those principles to improve your presentations

[Do]

Principles for making compelling PowerPoint presentations

Figure 3.12. As a general rule, two lines is about right for each bulleted entry. Try reading aloud the one on the top; if it's a mouthful for you, it's an earful for the audience.

7. Respect the *Rule of Four*: Use hierarchical organization, with no more than four items at each level.

If you have more than four concepts or items to present, organize them into categories and present each category in a separate slide. If you do this, first present a slide in which you list the categories in the order in which they will be presented, which will prepare the audience for what is to come.

Example: If you want to list 20 types of items commonly found in convenience stores, figure out how to organize them into four or fewer categories, then divide each category into four or fewer categories. As shown in Figure 3.13b, you might make an initial cut between food items and sundries (with perhaps 12 food items and 8 sundries); you then could divide the food items into four categories (such as candy, vegetables, baked goods, and dairy products) and the sundries into two categories.

[Don't]

Socks
Twix
Snickers
Sunglasses
Mars bars
Caps
Carrots
Peas
Squash
Insect repellant
Shoe laces
Shampoo
Bread
Hand lotion
Hotdog buns
Cake
Chap Stick
Donuts
Milk
Ice cream

Figure 3.13a. If you must present every single item, organize them into categories, each of which should contain no more than four entries. But before doing this, think again: Does the audience really need to see every one of those individual items? Would selected examples help you to make your point more clearly?

[Do]

Food Items
- *Candy*
  - Twix
  - Snickers
  - Mars bars
- *Vegetables*
  - Carrots
  - Peas
  - Squash
- *Baked goods*
  - Bread
  - Hotdog buns
  - Cake
  - Donuts
- *Diary*
  - Milk
  - Ice cream

Sundries
- *Wearables*
  - Sunglasses
  - Socks
  - Shoe laces
  - Caps
- *Consumables*
  - Hand lotion
  - Chap Stick
  - Insect repellant
  - Shampoo

Figure 3.13b.

However, this advice is based on the assumption that each item is in fact important to the point you are making. If the individual items are not important (for instance, it could have been bagels instead of donuts), then all you need to do is present the categories and perhaps list selected examples of the items. Present no more and no less than the audience actually needs to know.

8. Use the same bullet symbol for each entry in a slide.

The standard "●" bullet is tried and true, and a wide range of viewers will have no problem understanding what it signifies. So as a general rule, you will not go wrong using the standard bullet symbol. But you can use another symbol for a bullet if you are certain that the audience will understand it, and that the symbol has special relevance to your presentation.

Example: A small dollar sign might be appropriate if you are listing sources of income.

- Don't change the shape of the bullets in your presentation unless you want to signal a change in category. Viewers will assume that any visual change—replacing round bullets with arrows, for example—is intended to highlight something new about the material you are presenting.

- However, if you do use different symbols for bullets for different classes of information (e.g., income, number of workers, amount of rainfall in different locations), keep in mind that the audience will need a moment to understand that the different symbols function as bullets.

[Don't]

• Vary bullets arbitrarily for the same class of items
> Vary bullets arbitrarily for the same class of items
~ Vary bullets arbitrarily for the same class of items
# Vary bullets arbitrarily for the same class of items

[Do]

- Use the same bullet for the same classes of items
- Don't change the shape of bullets arbitrarily
- Use a standard bullet unless there's a good reason not to
- If you use a novel bullet, ensure that it is appropriate

Always remember that your job is to communicate clearly and compellingly, keeping the audience's attention throughout your presentation; so consider whether it's worth including additional features that will require the audience to work harder. As a general rule, my advice is to avoid taxing the audience's natural limitations if at all possible, particularly with formatting.

Labels: Guidelines for Making Words and Objects Work Together

In many cases you will pair words with graphics or tables, such as charts, graphs, diagrams, or illustrations (including photographs). Some labels are easier to grasp than others, and some may be effective only if the viewers are engaged and thinking through the implications of what you say. Consider this (probably apocryphal story): A member of the Massachusetts Senate, which has sometimes been a site of high drama and strong words, lost his temper after watching his colleagues vote for an obvious piece of Pure Pork, and, pounding on his desk, shouted, "Half of this Senate is made up of idiots and corrupt politicians!" The chamber broke into an angry babble, and the Senators demanded that he retract this slander. Looking up toward the ceiling, as if beseeching divine guidance, the offending senator spoke in a softer voice: "I retract my previous statement. Half of this august body is NOT made up of idiots and corrupt politicians!" The first statement required no work to grasp its meaning, and the second one is (moderately) amusing because it is clear only after you put a tiny bit of work into decoding its implications. (I've heard this one in various forms, and adapted this version from http://jokes.comedycentral.com—which, in spite of the source, is no joke.) But if the audience is distracted or otherwise not motivated to pay attention, they may not draw out even simple implications. Don't risk it; serve up your offering as transparently and directly as possible.

1. Use terms that convey the appropriate denotations and connotations.

Words not only have *denotations* (direct meanings) but also *connotations* (indirect meanings), and you need to make sure that both are appropriate for your particular audience and for the point you want to make.

Example: You could describe a new idea as "innovative" (which has a positive connotation) or "new-fangled" (which has a negative connotation). If you are promoting the idea, the connotation of the second term would get in your way.

2. Label every graphic, table, and their information-bearing parts—unless the identity is self-evident.

No matter how fascinating you and your material may be, some members of the audience may allow their attention to wander from time to time. In fact, if your presentation is unusually stimulating, you may be rewarded by sparking the audience to consider implications and applications—which will inadvertently result in their missing snatches of what you have to say. As a general rule, label everything important, so that viewers can read what they failed to hear.

Example: If you have a photo of a historical personage, provide the name; if you have a map of Beijing, label it.

- Don't label symbols or material that are extremely familiar to the audience, such as a picture of the current president or a very famous landmark. If information is not needed, don't supply it (if you do, the audience might think that you are talking down to them).

3. Group labels with the appropriate elements of the display.

To ensure that each label is clearly associated with the element it identifies, keep in mind the various aspects of the Principle of Perceptual Organization, and beware of the following:

- Viewers will group labels that are placed near a graphic with that graphic; so place labels only near graphics to which they apply.
- Viewers will see labels that seem to flow from the end of a line as part of that line, and hence assume that the labels apply to that line. Avoid this arrangement unless the label does in fact apply.
- Viewers will group labels that are the same color as an element with that element (e.g., green labels with green wedges, blue labels with blue wedges, and so forth). Such grouping is confusing when the labels don't in fact apply to the elements with the same color, so avoid doing this.

4. Label content elements directly.

Place labels immediately next to or within the corresponding content elements, such as objects, bars, lines, or wedges of a pie chart. If it is not possible to place a label near what it identifies, or if placing such a label creates clutter, use a key. But use a key only as a last resort because a key will force the audience to memorize and search, and both tasks require effort that the audience may not be willing to expend.

5. Make words in the same label close together and typographically similar.

Take advantage of the Principle of Perceptual Organization by ensuring that words in the same label do cohere as a unit and don't group individually with other parts of the display.

- Place the letters, numbers, and words in a label relatively close to one another.
- Make the letters, numbers, and words in a label the same size, color, and brightness.
- Be certain that labels are not so close to one another that they group together improperly.

Example: These words group properly . . . These words . . . do . . . NOT.

6. Use more salient labels to identify more important parts of the display.

As we know from the Principle of Salience, our visual systems reflexively attend to extreme values and large differences in line length and width, shading, color, motion, and other visual properties. As a result, we notice heavier lines before thinner ones, lines that contrast with the color of their backgrounds before lines that blend with their backgrounds, brighter colors before dimmer ones, thicker bars before thinner ones, and moving objects before stationary ones. In accordance with the Principle of Salience, we should

- make the labels for the most important parts of the display more salient (for example, by having them be relatively large, bold, a striking color, or moving in from the side).
- make the labels for increasingly smaller details, or more fine-grained components, decreasingly less salient.

7. Use hierarchical labeling.

Take a look at the two panels, labeled "Don't" and "Do" in the illustration below. Did Hillsborough or Pinellas have greater sales of single-family homes in 1991? This is relatively easy to determine from Do, but much harder from Don't. When the label refers to two or more dimensions, use a hierarchical labeling system. A hierarchical labeling system is one in which some labels apply to groups of other labels; the "higher in the hierarchy" the label is, the larger it should be. As illustrated in Figure 3.15b, "Do," using a hierarchical labeling system achieves three goals. This scheme

- eliminates the need for a key, which taxes the processing capacities of the audience members because it requires them to search for the corresponding parts of the display;
- specifies each of the relevant dimensions separately, which helps the audience to make specific comparisons;
- removes redundant labels, which clutter a display, thereby making it more difficult to pick out the other elements.

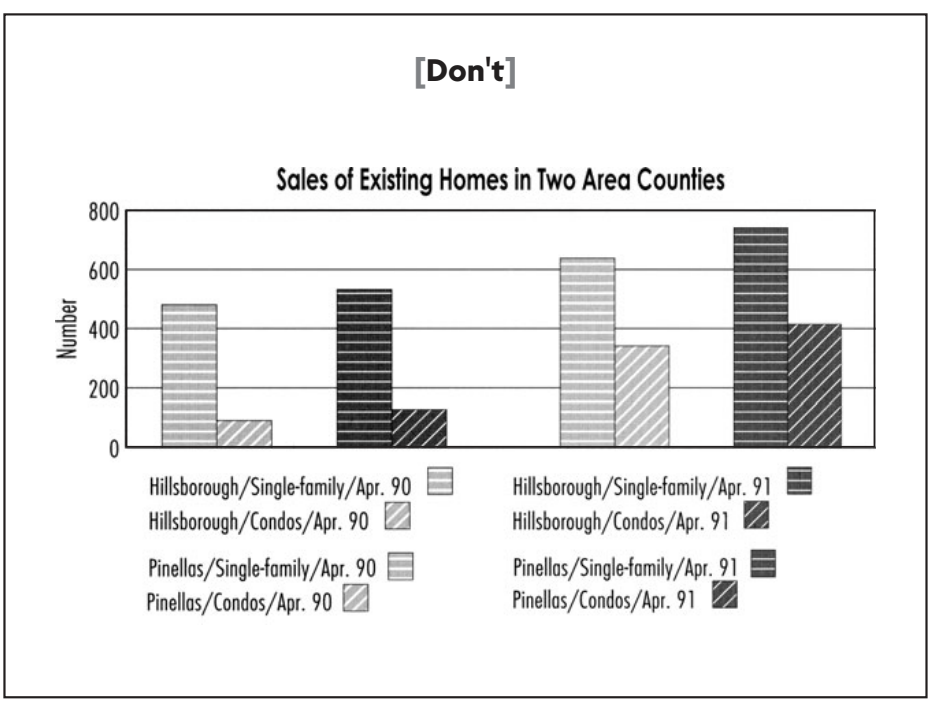

Figure 3.15a.

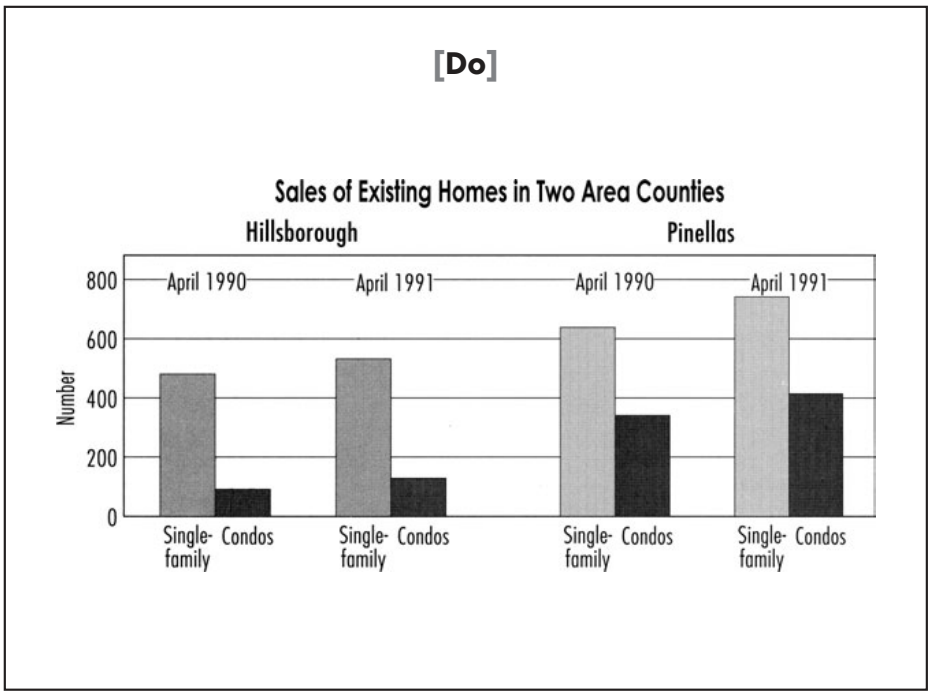

Figure 3.15b. Not only is the display less cluttered when direct labels are used, but also the bars are easier to compare along specific dimensions (dwelling type, county, year) when the labels specify the dimensions separately.

8. Use the same size and font for labels that identify the same types of components.

When we see a difference, we expect it to mean something and are confused if it doesn't. Labels for each of the same type of component (wedges, bars, etc.) should use the same font and be the same size. Entities of equal importance should be labeled with the same font and be the same size.

Example: If you have two pictures of former CEOs and the sizes of their names or their photos are different, the audience will draw the obvious conclusion about your level of regard for each.

9. Use the same terminology in labels, as well as in the surrounding text and spoken words.

Using different terms in a display, in text, and in what you say aloud may lead the audience to wonder if you mean different things. Attempting to distinguish these differences will definitely tax the cognitive capacities of your audience.

Example: Don't label birds "birds" on a slide while referring to "fowl" in text or in your spoken remarks.

Titles: 4 Simple Guidelines

The psychological principles will also help you to create titles for your slides.

1. Title a slide to focus attention on the most important information.

I recently ran across the following quote, attributed to the late Jerry Garcia (of the Grateful Dead rock band): "I read somewhere that 77 percent of all the mentally ill live in poverty. Actually, I'm more intrigued by the 23 percent who are apparently doing quite well for themselves." He was right! To convey a different message, someone could have said, "Twenty-three percent of the mentally ill do well financially." Until I read Mr. Garcia's remark, I hadn't ever looked at it that way. In general, the same information can be presented in many different ways; different aspects can be highlighted over other aspects. You need to decide what's the conceptual foreground and what's the conceptual background. The title of a slide tells the reader what is the foreground (and, if relevant, what is the background).

- When formulating a title, ask yourself: What material do you want to stress? Does it apply in particular to a specific time (year, month) and place (country, region of a state)?
- Include only the most relevant information in the title. Define the foreground, and only mention the background if it is immediately relevant.
- Mention what's most important first.

Example: "Effects of mental illness on earnings of men versus women" puts "mental illness" in the foreground and gender in the background, whereas "Effects of gender on earnings in the mentally ill" puts gender in the foreground and mental illness in the background.

2. Make the title typographically distinct.

The title should be the most salient element of the display. Use the most salient font for the title, to catch the eye immediately.

3. Present the title of a complex slide before the content elements are clearly visible.

Leading with the title will force your audience to pay attention to it, which will give them a framework for understanding the rest of your material more easily. So, if you have a very elaborate slide, present the title first, with everything else grayed out (i.e., in a very low-contrast font).

4. Center the title at the top of the slide.

Audiences are used to seeing titles at the tops of documents, so that is where they expect the title to be on your slides. If it is off center, the audience may group it with one part of the slide, rather than with the whole slide.

If you place the title anywhere but the center of the top—for example, in the left margin—you must follow this convention through your entire presentation, creating a distinctive but consistent design. Viewers will grow familiar with this design and will quickly come to group the title properly. However, an unconventional style will initially make viewers work harder than a conventional one will.

Tables: 8 Legs to Stand On

Tables include lists of numbers or words organized into rows and columns. Here are eight recommendations that will help you make your tables more useful.

1. Use a table of numbers only when specific values are important.

Use a table of numbers only if it is important to convey specific values, as opposed to differences among values. You will be better off using a graph if your goal is simply to show that one entity is greater than another.

2. Don't use a table if trends or interactions are important.

If your goal is to present a specific trend (such as increased revenues over time) or a difference between trends (such as increased rev-

enues over time in one part of the world with no such increases in another), use a line graph. Using a table would force viewers to compute and compare differences among differences of numbers, which requires work. As always, don't make viewers struggle to grasp your message.

3. Show complex tables to emphasize that the data are complex or to show a data point in context.

Just as Abe Lincoln would never have been silly enough to try to use a PowerPoint presentation to deliver the Gettysburg address, you should not even think of using such a presentation as a substitute for a research article. PowerPoint presentations and research articles just aren't the same things; they have different purposes. In an article, you need to report all the details (and thus a complex table is appropriate). In a presentation, you need to show how material leads to a certain conclusion and shouldn't ask viewers to spend even 5 minutes sitting silently in their seats, pondering the various differences among numbers in a table you project.

The only time to present a very complex table is when you want to make the point that

- there are a lot of data arranged in orderly ways, or
- to highlight a single value (e.g., by putting it in red) and show it in relation to the table as a whole. In this case, consider including a follow-on slide in which you "zoom in" on the critical value and its immediate surrounds.

If you have so many entries that they must be very small in order to fit on a slide, this is a painfully clear signal that your table is not going to be very useful for other purposes. You will probably be better off splitting the table into two or more slides. If this won't work, consider using a bar graph to present averages (with some measure of variation, such as the standard error of the mean, around each average value) rather than a table.

4. Present only the information needed to make your point.

A friend of mine sent me an e-mail about the following incident (I'm not sure whether this was supposed to be accurate or a joke, but either way it makes my point): A high school chemistry teacher was annoyed at her class for not mastering the Periodic Table; the students didn't seem to understand why the table is important and hadn't put much effort into memorizing it by rote. In a mild pique,

the teacher told her students that when she was their age, she knew all the elements in the table backward and forward. One soft voice from the back of the room muttered in reply, "Sure, but in those days there were only a few to learn." She couldn't help but smile, but if you show a very complex table without providing a very good reason for it, your audience probably won't. Present no more information than you need to make your point (and no less). Keep in mind that unnecessary distinctions can confuse the audience just as quickly as unnecessary data.

Example: If you are not interested in monthly fluctuations, don't present values broken down by month.

5. Organize a table to emphasize more important distinctions.

Use formatting to emphasize the most important aspects of the presented information.

Example: If you are presenting data from men and women in two countries, and the gender difference is more important for your point, you would organize the table as on the left of Table 3.1. If, on the other hand, country is more important, then you would organize the data as on the right. Note again the use of hierarchical labeling.

Table 3.1

	Men		Women	
	USA	Canada	USA	Canada
Measure 1	14	15	12	10
Measure 2	10	18	9	10
Measure 3	12	14	14	13
Measure 4	10	12	10	12

	USA		Canada	
	Men	Women	Men	Women
Measure 1	14	12	15	10
Measure 2	10	9	18	10
Measure 3	12	14	14	13
Measure 4	10	10	12	12

6. In complex tables, include summary statistics and make them salient.

If you need to present more than two columns and two rows of numbers, include summary statistics (e.g., means or sums, as appropriate for your message) at the bottoms of the columns and right-hand sides of the rows. (In the trade, these summary statistics are known as "marginal values.")

- Summary statistics should be heavier, larger, or more salient in some other way (e.g., having a brighter color) than the entries in the table.

7. In complex tables, include grid lines.

If you need to present more than two columns and two rows, use grid lines. Grid lines will not only help viewers to organize the table but will also guide their eyes from the labels of rows and columns to the specific entries.

8. If the entries are labeled with pictures or icons, ensure that they evoke the appropriate concept.

People are more likely to name objects consistently if a picture illustrates a "prototypical example" (i.e., an average member) of the category. If you want to use graphics to label entries in a table, ensure that the audience will easily associate the graphics with the appropriate entities.

Example: If you want to illustrate different kinds of animals that are kept as pets, an average-looking mutt is more likely to be labeled "dog" than a St. Bernard would be.

The Key: Factors That Make It Work

A key has two components: (1) patches of a graphic or table that correspond to individual components; and (2) labels that identify these patches. These patches typically consist of squares that contain different colors, texture patterns, or segments of dashed versus solid lines.

1. Use a key when direct labels cannot be used.

A key requires the viewer to memorize an association between labels and portions of the graphic or table, and then to find those portions in order to identify them. So you should avoid using a key unless absolutely necessary. There are two exceptions to this recommendation, both of which are illustrated in Figure 3.16, "Do":

- Use a key when there are so many wedges, objects, or segments in a small space that is impossible to label them directly because the labels would not group properly or they would have to be so small as to be illegible.
- Use a key when the same entities appear in more than two parts of a graphic or table, so a key will reduce clutter and the difficulty of searching for labels.

That said, if you need a key, consider whether you want to show the graphic or table in your presentation; the display may simply be too complicated to be grasped easily in this context. If you do need it:

- consider building up the display a part at a time, so that the key becomes a way to keep track of the separate parts you have shown.
- be sure to point out explicitly the relation between parts of the key and the corresponding parts of the display.

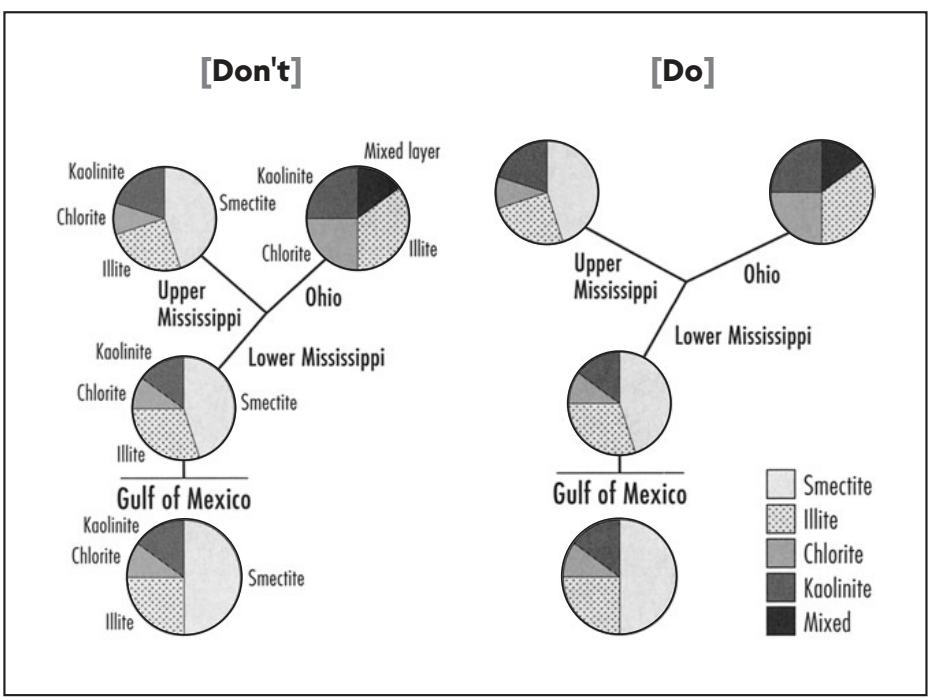

Figure 3.16. A key is sometimes appropriate, particularly when it reduces clutter and allows the structure of the display itself to be more prominent.

2. Place a key at the top right of a single panel or centered over multiple panels.

In a single-panel display, the key is by convention positioned at the top right; in a multipanel display, which contains more than one

variant of the same graphic (such as graphs or diagrams), it is centered at the top, directly beneath the title. Viewers familiar with graphics will expect to find the key in one of these locations. If aesthetic or other considerations lead you to put the key in an unconventional location (such as in the example you just saw), realize that the viewers may have to search for it, which will tax their limited processing capacity. However, if you consistently place the key in the same location in a number of displays, the viewers will soon learn where to find it.

3. Ensure that labels and patches are easy to tell apart.

Ensure that each label and patch in the key is discriminable from the back of the room and when the lights are turned on. To ensure that they are discriminable, you may need to make patches larger than they would be if you were publishing the display in an article. Furthermore, if you use different textures in the patches and in the corresponding portions of the display, keep in mind that a PowerPoint slide does not have the resolution of a published figure (and thus relatively coarse patterns are appropriate for PowerPoint).

In addition, if you want the audience to make specific comparisons, it's a good idea to point out explicitly the correspondence between the relevant patches in the key and the portion or portions of the display they label.

4. Ensure that corresponding labels and patches form perceptual groups.

Associate the label and the patch by making sure that they are closer to each other than to any other part of the key; the proximity will then group the elements properly.

5. Use the same order for patches and their corresponding content elements.

The patches in the key should be presented in the same order as the corresponding content elements (bars, icons, etc.) in the display itself.

Example: If the labeled patches in the key refer to elements in a row, as in "Do," the top element should refer to the leftmost content element in the display, the second from the top element should refer to the content element second from the left, and so forth. These guidelines are followed in "Do" of Figure 3.17.

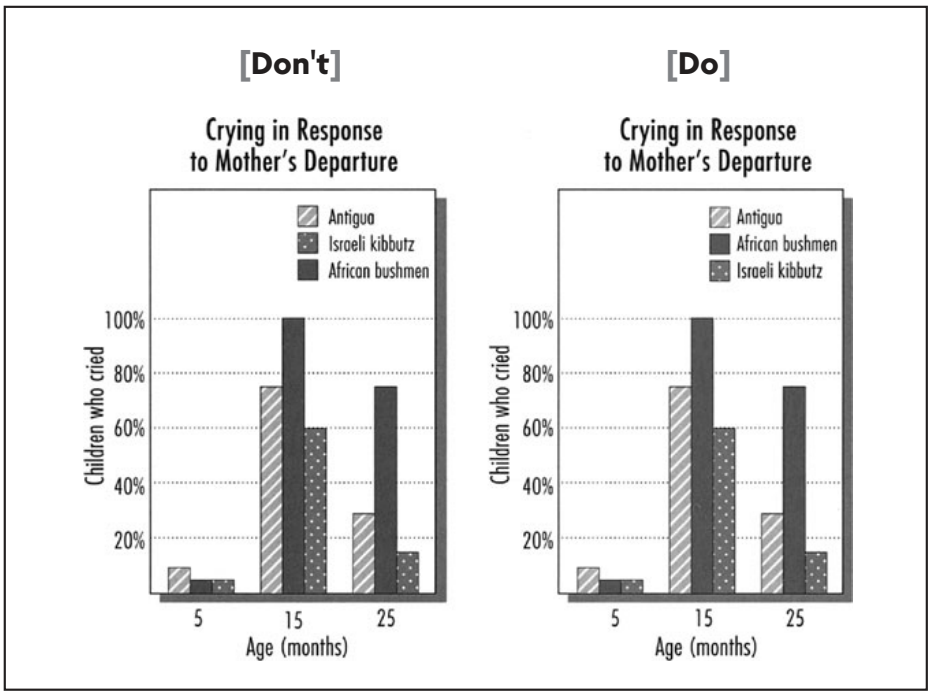

Figure 3.17. Viewers will have to search for corresponding bars if the order of the key does not match the order of the content elements.

From the Principles to the Point

These recommendations are based on the same psychological principles discussed earlier. Let's revisit them and note which ones contributed the most to each of the major recommendations I offer.

Principle of Relevance: Communication is most effective when neither too much nor too little information is presented. (We understand and remember a message more easily when the right amount of detail is used to make the point.)

- Present a quick overview of a list, then gray out all but the first entry.
- Ensure that only the relevant words are easily distinguishable from the background.
- Present items individually, highlighting only one at a time as you go from top to bottom.
- Use bullets only for topic sentences or specific cases.
- Title a slide to focus attention on the most important information.
- Use a table of numbers only when specific values are important.
- Present only the information needed to make your point.

Principle of Appropriate Knowledge: Communication requires prior knowledge of pertinent concepts, jargon, and symbols. (We understand and remember a message more easily if it connects to what we already know.)

- Use standard fonts.
- Use terms that convey the appropriate denotations and connotations for your audience and the point you want to make.
- Center the title at the top of the slide.
- Place a key at the top right of a single panel or centered over multiple panels.

Principle of Salience: Attention is drawn to large perceptible differences. (Big relative differences grab attention.)

- Use more salient labels to label more important components of the display, and use less salient labels for details or smaller components.
- Make the title typographically distinct.

Principle of Discriminability: Two properties must differ by a large enough proportion or they will not be distinguished. (We need contrast to distinguish shapes, colors, or positions from each other and from the background.)

- Avoid all uppercase, all italic, or all bold typefaces.
- Don't underline (use bold, color, or italics for emphasis).
- Use fonts that are easy to read.
- Use visually simple fonts.
- Ensure that words are large enough to be read easily.
- Ensure that viewers can easily discriminate text from the background.
- Use either a serif or a sans serif font, but do not mix and mingle them arbitrarily.
- If entries in a table are too small to be read, show the table in two or more slides.
- Ensure that labels and patches are easy to tell apart.

Principle of Perceptual Organization: People automatically group elements into units, which they then attend to and remember. (These groups are easier to see and remember than the isolated components would be.)

- Group labels with the appropriate elements of the display.
- Label content elements directly whenever space permits.
- Make words in the same label close together and typographically similar.
- In complex tables, include grid lines.

- Organize a table to emphasize more important distinctions.
- Ensure that corresponding labels and patches form perceptual groups.

Principle of Compatibility: A message is easiest to understand if its form is compatible with its meaning. (For better or worse, the mind tends to judge a book by its cover.)

- If entries are labeled with pictures or icons, ensure that they evoke the appropriate concept.
- Discuss only material that is closely related to what is currently being shown on a slide.
- Don't use a table if trends or interactions are important (usually, use a line graph).

Principle of Informative Changes: People expect changes in properties to carry information. (And we expect every necessary piece of information to be indicated by a change in a perceptible property.)

- Use different colors only for emphasis or to specify different classes of information.
- Use different fonts only for emphasis or to specify different classes of information.
- Use the same bullet symbol for each entry in a list of similar items; change bullet symbols only to indicate a change in the nature of the entry.
- Label every graphic, table, and their information-bearing parts—unless the identity is self-evident.
- Use the same size and font for labels that identify the same types of components.
- Use the same terminology in labels as well as in the surrounding text and spoken words.
- Use the same order for patches in a key and their corresponding content elements in a graphic or table.

Principle of Capacity Limitations: People have a limited capacity to retain and to process information, and so will not understand a message if too much information must be retained or processed. (From a communicative point of view, less can be more!)

- Present parts of complex displays individually, growing the whole from the parts.
- Give the audience time to read each bulleted item.
- Use no more than two lines per bulleted entry.
- Show no more than four bulleted items in a single list.

- Respect the *Rule of Four*: Use hierarchical organization, with no more than four items at each level.
- Use hierarchical labeling.
- Present the title of a complex slide before the easily visible content elements.
- Show complex tables to emphasize that the data are complex or to show a data point in context.
- In complex tables, include summary statistics and make them salient.
- Use a key when direct labels cannot be used.

If you master the principles, you won't need to memorize every one of the recommendations any more than you need to memorize every sentence you might say today. Just as you can generate sentences on the basis of your (unconscious) knowledge of grammar and words, with practice you can learn to generate good PowerPoint presentations on the basis of your knowledge of human mental processes.